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APPARATUS AND METHOD FOR MANUFACTURING SECONDARY BATTERY AND SECONDARY BATTERY MANUFACTURED THEREBY

Abstract

Disclosed are a secondary battery manufacturing apparatus and method capable of preventing wrinkles and folding of a substrate tab during manufacture of an electrode plate and preventing rupture of the electrode plate during travel or winding of the electrode plate and a secondary battery manufactured thereby. The manufacturing apparatus includes an unwinding unit unwinding an electrode plate wound on a first roller, the electrode plate including a coated portion in which an active material is coated on a substrate portion and an uncoated portion in which an active material is not coated on the substrate portion, a fold prevention portion forming unit forming a fold prevention portion on the uncoated portion of the electrode plate, a notching unit notching the uncoated portion including the fold prevention portion to form a substrate tab, and a winding unit winding the electrode plate including the substrate tab on a second roller.

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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATIONS

[0001] This present application claims priority to and the benefit under 35 U.S.C. § 119(a)-(d) of Korean Patent Application No. 10-2024-0032977, filed on Mar. 8, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

FIELD

[0002] Aspects of embodiments of the present disclosure relate to an apparatus and method for manufacturing a secondary battery and a secondary battery manufactured thereby.

BACKGROUND

[0003] Unlike primary batteries that are not designed to be (re) charged, secondary (or rechargeable) batteries are batteries that are designed to be discharged and recharged. Low-capacity secondary batteries are used in portable, small electronic devices, such as smart phones, feature phones, notebook computers, digital cameras, and camcorders, while large-capacity secondary batteries are widely used as power sources for driving motors in hybrid vehicles and electric vehicles and for storing power (e.g., home and/or utility scale power storage). A secondary battery generally includes an electrode assembly composed of a positive electrode and a negative electrode, a case accommodating the same, and electrode terminals connected to the electrode assembly.

[0004] The above information disclosed in this Background section is for enhancement of understanding of the background of the present disclosure, and therefore, it may contain information that does not constitute related (or prior) art.

SUMMARY

[0005] According to an aspect of one or more embodiments of the present disclosure, a secondary battery manufacturing apparatus and method capable of preventing wrinkles and folding of a substrate tab during a process of manufacturing an electrode plate of a secondary battery and preventing rupture of the electrode plate during travel or winding of the electrode plate and a secondary battery manufactured thereby is provided.

[0006] These and other aspects and features of the present disclosure will be described in or will be apparent from the following description of embodiments of the present disclosure.

[0007] An apparatus for manufacturing a secondary battery according to an embodiment of the present disclosure includes an unwinding unit configured to unwind an electrode plate wound on a first roller, the electrode plate including an active material coated portion in which an active material is coated on a substrate portion and an uncoated portion in which the active material is not coated on the substrate portion, a fold prevention portion forming unit configured to form a fold prevention portion on the uncoated portion of the electrode plate, a notching unit configured to notch the uncoated portion including the fold prevention portion formed thereon to form a substrate tab, and a winding unit configured to wind the electrode plate including the substrate tab formed thereon on a second roller.

[0008] In one or more embodiments, the fold prevention portion forming unit may include a

driving roller including a circular area and an uneven area, the circular area and the uneven area disposed on the same axis while being spaced apart from each other, and a driven roller configured to rotate in close proximity to the uneven area of the driving roller. The electrode plate may travel between the driving roller and the driven roller such that the active material coated portion of the electrode plate travels in contact with the circular area of the driving roller and the uncoated portion of the electrode plate travels in contact with the uneven area of the driving roller and the driven roller, such that the fold prevention portion may be imprinted on the uncoated portion.

[0009] In one or more embodiments, the elastic modulus of the uneven area may be greater than the elastic modulus of the driven roller.

[0010] In one or more embodiments, the uncoated portion may include a first surface and a second surface opposite the first surface, the uneven area may contact the first surface of the uncoated portion, the driven roller may contact the second surface of the uncoated portion, and the rotational axis of the uneven area and the rotational axis of the driven roller may overlap each other in a vertical direction.

[0011] In one or more embodiments, the fold prevention portion may comprise a pattern including a plurality of straight lines extending in a direction perpendicular to the longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

[0012] In one or more embodiments, the fold prevention portion may comprise a pattern including a plurality of straight lines extending in a direction parallel to the longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

[0013] In one or more embodiments, the fold prevention portion may comprise a pattern including a plurality of oblique lines extending at an angle with respect to the longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

[0014] In one or more embodiments, the fold prevention portion may comprise a pattern including a plurality of straight lines extending in a direction perpendicular to the longitudinal direction of the uncoated portion and at least one straight line extending in a direction parallel to the longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

[0015] In one or more embodiments, the uneven area of the driving roller may have a depth of 0.1 mm to 2 mm.

[0016] In order to accomplish the above objective, a method of manufacturing a secondary battery according to an embodiment of the present disclosure includes unwinding an electrode plate wound on a first roller, the electrode plate including an active material coated portion in which an active material is coated on a substrate portion and an uncoated portion in which an active material is not coated on the substrate portion, forming a fold prevention portion on the uncoated portion of the electrode plate, notching the uncoated portion including the fold prevention portion formed thereon to form a substrate tab, and winding the electrode plate including the substrate tab formed thereon on a second roller.

[0017] In one or more embodiments, in the forming a fold prevention portion, a driving roller including a circular area and an uneven area, the circular area and the uneven area disposed on the same axis while being spaced apart from each other, and a driven roller configured to rotate in close proximity to the uneven area of the driving roller may be provided. The electrode plate may travel between the driving roller and the driven roller such that the active material coated portion of the electrode plate travels in contact with the circular area of the driving roller and the uncoated portion of the electrode plate travels in contact with the uneven area of the driving roller and the driven roller, such that the fold prevention portion may be imprinted on the uncoated portion.

[0018] In one or more embodiments, the elastic modulus of the uneven area may be greater than the elastic modulus of the driven roller.

[0019] In one or more embodiments, the uncoated portion may include a first surface and a second surface opposite the first surface, the uneven area may contact the first surface of the uncoated portion, the driven roller may contact the second surface of the uncoated portion, and the rotational

axis of the uneven area and the rotational axis of the driven roller may overlap each other in a vertical direction.

[0020] In one or more embodiments, in the forming a fold prevention portion, the fold prevention portion may be formed in a pattern including a plurality of straight lines extending in a direction perpendicular to the longitudinal direction of the uncoated portion.

[0021] In one or more embodiments, in the forming a fold prevention portion, the fold prevention portion may be formed in a pattern including a plurality of straight lines extending in a direction parallel to the longitudinal direction of the uncoated portion.

[0022] In one or more embodiments, in the forming a fold prevention portion, the fold prevention portion may be formed in a pattern including a plurality of oblique lines extending at an angle with respect to the longitudinal direction of the uncoated portion.

[0023] In one or more embodiments, in the forming a fold prevention portion, the fold prevention portion may be formed in a pattern including a plurality of straight lines extending in a direction perpendicular to the longitudinal direction of the uncoated portion and at least one straight line extending in a direction parallel to the longitudinal direction of the uncoated portion.

[0024] In one or more embodiments, the uneven area of the driving roller may be formed to have a depth of 0.1 mm to 2 mm.

[0025] In order to accomplish the above objective, a secondary battery according to an embodiment of the present disclosure is manufactured by the method described above.

[0026] According to some embodiments, an apparatus for manufacturing a secondary battery comprises a fold prevention portion forming unit configured to form a fold prevention portion on an uncoated portion of an electrode plate; and a notching unit configured to notch the uncoated portion comprising the fold prevention portion formed thereon to form a substrate tab.

[0027] However, aspects and features of the present disclosure are not limited to those described above, and other aspects and features not mentioned will be clearly understood by a person skilled in the art from the detailed description, described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The following drawings attached to this specification illustrate embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed as being limited to the drawings:

[0029] FIG. 1A is an exploded perspective view showing an exemplary secondary battery according to the present disclosure according to some embodiments;

[0030] FIG. 1B is a partial exploded perspective view showing an exemplary electrode assembly accommodated in the exemplary secondary battery according to some embodiments;

[0031] FIG. 2A is a schematic diagram showing an exemplary apparatus for manufacturing a secondary battery according to some embodiments;

[0032] FIG. 2B is a flowchart showing an exemplary method of manufacturing a secondary battery according to some embodiments;

[0033] FIG. 3A is a perspective view showing an exemplary fold prevention portion forming unit of the exemplary apparatus for manufacturing a secondary battery according to some embodiments;

[0034] FIG. 3B is a front view showing a driving roller of the exemplary fold prevention portion forming unit according to some embodiments;

[0035] FIG. 3C illustrates a side view and a partial enlarged view showing the driving roller of the exemplary fold prevention portion forming unit according to some embodiments;

[0036] FIG. 4 is a schematic diagram showing the operation of the exemplary apparatus for

manufacturing a secondary battery according to some embodiments;

[0037] FIGS. 5A to 5D are schematic diagrams showing various forms of a fold prevention portion of an electrode plate and the driving roller of the exemplary apparatus for manufacturing a secondary battery according to some embodiments;

[0038] FIGS. 6A and 6B are perspective views showing a battery pack including the exemplary secondary battery according to some embodiments; and

[0039] FIGS. 7A and 7B are, respectively, a perspective view and a side view showing a vehicle body and a vehicle including the exemplary battery pack according to some embodiments.

DETAILED DESCRIPTION

[0040] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in the present specification and claims are not to be limitedly interpreted as general or dictionary meanings and should be interpreted as meanings and concepts that are consistent with the technical idea of the present disclosure on the basis of the principle that an inventor can be his/her own lexicographer to appropriately define concepts of terms to describe his/her invention in the best way.

[0041] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical spirit, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0042] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.

[0043] In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B and C,” “at least one of A, B or C,” “at least one selected from a group of A, B and C,” or “at least one selected from among A, B and C” are used to designate a list of elements A, B and C, the phrase may refer to any and all suitable combinations or a subset of A, B and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0044] It will be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0045] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

[0046] The terminology used herein is for the purpose of describing embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0047] Also, any numerical range disclosed and/or recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein. All such ranges are intended to be inherently described in this specification such that amending to expressly recite any such subranges would comply with the requirements of 35 U.S.C. § 112(a) and 35 U.S.C. § 132(a).

[0048] References to two compared elements, features, etc. as being “the same” may mean that they are “substantially the same”. Thus, the phrase “substantially the same” may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.

[0049] Throughout the specification, unless otherwise stated, each element may be singular or plural.

[0050] Arranging an arbitrary element “above (or below)” or “on (under)” another element may mean that the arbitrary element may be disposed in contact with the upper (or lower) surface of the element, and another element may also be interposed between the element and the arbitrary element disposed on (or under) the element.

[0051] In addition, it will be understood that when a component is referred to as being “linked,” “coupled,” or “connected” to another component, the elements may be directly “coupled,” “linked” or “connected” to each other, or another component may be “interposed” between the components”.

[0052] Throughout the specification, when “A and/or B” is stated, it means A, B or A and B, unless otherwise stated. That is, “and/or” includes any or all combinations of a plurality of items enumerated. When “C to D” is stated, it means C or more and D or less, unless otherwise specified.

[0053] FIG. 1A is an exploded perspective view showing an exemplary secondary battery **100** according to some embodiments of the present disclosure, and FIG. 1B is a partial exploded

perspective view showing an exemplary electrode assembly **110** accommodated in the exemplary secondary battery **100** according to some embodiments of the present disclosure. As shown in FIGS. **1A** and **1B**, the secondary battery **100** may include an electrode assembly **110**, a first lead tab **121**, a second lead tab **122**, and a case **130**.

[0054] The electrode assembly **110** may be formed by winding or stacking a stack of a first electrode plate **111**, a second electrode plate **112**, and a separator **113** interposed between the first and second electrode plates **111** and **112**. The first and second electrode plates **111** and **112** and the separator **113** may be formed in the shape of a thin plate or a thin film. In some embodiments, the electrode assembly **110** is formed by winding the stack, and the winding axis thereof may be parallel to the longitudinal direction (Y direction) of the case **130**. In some embodiments, the electrode assembly **110** may be of a stacked type rather than a wound type. In some embodiments, the electrode assembly **110** may be a Z stack electrode assembly in which the separator **113** is bent in a Z-shape, and a positive electrode plate and a negative electrode plate are respectively inserted into both sides of the separator **113**. In some embodiments, one or more electrode assemblies **110** may be accommodated in the case **130** in a state of being stacked such that the long side surfaces thereof are adjacent to each other. In one or more embodiments, the first electrode plate **111** of the electrode assembly **110** may serve as a negative electrode, and the second electrode plate **112** of the electrode assembly **110** may serve as a positive electrode. In other embodiments, the reverse may also be possible.

[0055] The first electrode plate **111** may be formed by coating a first electrode active material (e.g., a negative electrode active material) **1112**, such as graphite or carbon, on a first electrode current collector **1111**, which is formed as a metal foil made of copper, a copper alloy, nickel, and/or a nickel alloy, and may include a plurality of first electrode tabs **1113**, which are areas not coated with the first electrode active material **1112**. In one or more embodiments, the first electrode current collector **1111** may be referred to as a first electrode substrate, and the first electrode tab **1113** may be referred to as a first substrate tab or a first uncoated portion tab. The first electrode tab **1113** may serve as a passage for flow of current between the first electrode plate **111** and the first lead tab **121**. In one or more embodiments, the first electrode tab **1113** may be formed in advance through notching or cutting so as to protrude in one direction when manufacturing the first electrode plate **111**, or may be formed so as to protrude farther in one direction than the separator **113** without a separate cutting process. The first electrode tab **1113** may include a plurality of fold prevention portions **1114** formed in the longitudinal direction of the first electrode tab **1113**. In one or more embodiments, the fold prevention portion **1114** may include a V-shaped or U-shaped uneven pattern. The structure of the fold prevention portion **1114** and a method of manufacturing the same will be described in detail later.

[0056] The second electrode plate **112** may be formed by coating a second electrode active material (e.g., a positive electrode active material) **1122**, such as a transition metal oxide, on a second electrode current collector **1121**, which is formed as a metal foil made of aluminum or an aluminum alloy, and may include a plurality of second electrode tabs **1123**, which are areas not coated with the second electrode active material **1122**. In one or more embodiments, the second electrode current collector **1121** may be referred to as a second electrode substrate, and the second electrode tab **1123** may be referred to as a second substrate tab or a second uncoated portion tab. The second electrode tab **1123** may serve as a passage for flow of current between the second electrode plate **112** and the second lead tab **122**. In one or more embodiments, the second electrode tab **1123** may be formed in advance through notching or cutting so as to protrude in one direction when manufacturing the second electrode plate **112**, or may be formed so as to protrude farther in one direction than the separator **113** without a separate cutting process. The second electrode tab **1123** may include a plurality of fold prevention portions **1124** formed in the longitudinal direction of the second electrode tab **1123**. In one or more embodiments, the fold prevention portion **1124** may include a V-shaped or U-shaped uneven pattern. The structure of the fold prevention portion

1124 and a method of manufacturing the same will be described in detail later.

[0057] In one or more embodiments, the first electrode tab **1113** may be located on the left side of the electrode assembly **110**, and the second electrode tab **1123** may be located on the right side of the electrode assembly **110**. For example, the left side and the right side may be defined on the basis of the secondary battery **100** shown in FIG. **1A** for convenience of description. The positions of the first and second electrode tabs **1113** and **1123** may be changed if the secondary battery **100** is rotated left and right or up and down.

[0058] The first electrode tab **1113** of the first electrode plate **111** and the second electrode tab **1123** of the second electrode plate **112** may be located on one end portion of the electrode assembly **110** so as to be spaced apart from each other. In one or more embodiments, the electrode assembly **110** may be accommodated in the case **130** together with an electrolyte. The plurality of first electrode tabs **1113** of the first electrode plate **111** and the plurality of second electrode tabs **1123** of the second electrode plate **112** may be welded and connected to the first lead tab **121** and the second lead tab **122**, respectively.

[0059] Each of the first lead tab **121** and the second lead tab **122** may extend from the inside of the case **130** to the outside of the case **130**. Similar to the first electrode tab **1113** and the second electrode tab **1123**, the first lead tab **121** and the second lead tab **122** may also be spaced apart from each other. In one or more embodiments, the first lead tab **121** may include a metal foil made of copper, a copper alloy, nickel, and/or a nickel alloy. In one or more embodiments, the second lead tab **122** may include a metal foil made of aluminum or an aluminum alloy. In one or more embodiments, a first sealing tape **1211** may be disposed on a portion of the first lead tab **121** that overlaps the case **130**, and a second sealing tape **1221** may be disposed on a portion of the second lead tab **122** that overlaps the case **130**.

[0060] The case **130** may accommodate both the electrode assembly **110** and the electrolyte. The case **130** may include or be referred to as a housing, an exterior body, a laminate exterior body, and/or a pouch. The case **130** may include a first case **130a**, which is substantially flat, and a second case **130c**, which is bent from the first case **130a** and has a recess **130b** formed therein. The second case **130c** may include three sealing portions **130d** provided around the recess **130b**.

[0061] The case **130** may be formed in a multilayered structure including a first insulating layer **131**, a second insulating layer **132**, and a metal layer **133** interposed between the first and second insulating layers **131** and **132**. In some embodiments, the case **130** may further include various other adhesive layers or functional layers. However, description thereof will be omitted so as not to obscure the gist of the disclosure. The first insulating layer **131** may function as an inner surface of the case **130**, and may be made of an insulative and thermally adhesive material. The first insulating layer **131** may be disposed on one surface of the metal layer **133** and may form the inner surface of the case **130** that faces the electrode assembly **110**. In one or more embodiments, the first insulating layer **131** may include polypropylene (PP) or casted polypropylene (CPP), which does not react with the electrolyte according to some embodiments. In some embodiments, the electrode assembly **110** is placed in the recess **130b** in the second case **130c** and is covered by the first case **130a**, and the first insulating layer **131** of the first case **130a** and the first insulating layer **131** of the second case **130c** may come into contact with each other. In some embodiments, thermal fusion is performed on the sealing portions **130d** wherein the first insulating layer **131** of the first case **130a** and the first insulating layer **131** of the second case **130c** may be bonded to each other such that the case **130** may be sealed.

[0062] The metal layer **133** may be interposed between the first insulating layer **131** and the second insulating layer **132**, and in some embodiments, may serve to block introduction of moisture and oxygen from the outside. In some embodiments, an electrolyte is accommodated in the case **130**, and the metal layer **133** may also serve, in such embodiments, to prevent leakage of the electrolyte to the outside. The metal layer **133** may also serve to maintain the mechanical strength of the case **130**. In one or more embodiments, the metal layer **133** may include aluminum, an aluminum alloy,

iron, and/or an iron alloy.

[0063] The second insulating layer **132** may function as an outer surface of the case **130**, and may serve to alleviate mechanical and chemical impact from external electronic devices. The second insulating layer **132** may be disposed on the other surface of the metal layer **133** and may form the outer surface of the case **130**. The second insulating layer **132** may include nylon, polyethylene terephthalate (PET), polybutylene terephthalate (PBT), and/or polybutylene naphthalate (PBN).

[0064] Although the pouch-type secondary battery **100** has been described, the electrode assembly **110** described above may also be applied in a similar or identical configuration or manner to a prismatic secondary battery or a cylindrical secondary battery.

[0065] Meanwhile, as the positive electrode active material, a compound capable of reversibly intercalating/deintercalating lithium (e.g., a lithiated intercalation compound) may be used. For example, at least one of a composite oxide of lithium and a metal selected from cobalt, manganese, nickel, and combinations thereof may be used.

[0066] The composite oxide may be a lithium transition metal composite oxide, and examples thereof may include a lithium nickel-based oxide, a lithium cobalt-based oxide, a lithium manganese-based oxide, a lithium iron phosphate-based compound, a cobalt-free nickel-manganese-based oxide, or a combination thereof.

[0067] As an example, a compound represented by any one of the following formulas may be used:

$\text{Li}_{0.90 \leq a \leq 1.8} \text{A}_{1-b} \text{X}_b \text{O}_{0.2-c} \text{D}_c$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.05$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{Mn}_{1-b} \text{X}_b \text{O}_{0.4-c} \text{D}_c$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.05$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{Ni}_{1-b} \text{CO}_{0.2-c} \text{X}_b \text{O}_{0.2-\alpha} \text{D}_\alpha$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.5$, $0 < \alpha < 2$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{Ni}_{1-b} \text{Mn}_b \text{X}_c \text{O}_{0.2-\alpha} \text{D}_\alpha$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.5$, $0 < \alpha < 2$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{Ni}_{0.90 \leq a \leq 1.8} \text{Co}_{0.90 \leq a \leq 1.8} \text{L}_{0.90 \leq a \leq 1.8}^{\text{sup.1}} \text{G}_{0.90 \leq a \leq 1.8} \text{O}_{0.2}$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.9$, $0 \leq c \leq 0.5$, $0 \leq d \leq 0.5$, $0 \leq e \leq 0.1$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{NiG}_{0.90 \leq a \leq 1.8} \text{O}_{0.2}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{CoG}_{0.90 \leq a \leq 1.8} \text{O}_{0.2}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{Mn}_{0.90 \leq a \leq 1.8} \text{G}_{0.90 \leq a \leq 1.8} \text{O}_{0.4}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$); $\text{Li}_{0.90 \leq a \leq 1.8} \text{Mn}_{0.90 \leq a \leq 1.8} \text{G}_{0.90 \leq a \leq 1.8} \text{PO}_{0.4}$ ($0.90 \leq a \leq 1.8$, $0 \leq g \leq 0.5$); $\text{Li}_{0.90 \leq a \leq 1.8} (3-f) \text{Fe}_{0.90 \leq a \leq 1.8}^{\text{sub.2}} (\text{PO}_{0.4})_{0.90 \leq a \leq 1.8}^{\text{sub.3}}$ ($0 \leq f \leq 2$); and $\text{Li}_{0.90 \leq a \leq 1.8} \text{aFePO}_{0.4}$ ($0.90 \leq a \leq 1.8$).

[0068] In the above formulas: A is Ni, Co, Mn, or a combination thereof; X is Al, Ni, Co, Mn, Cr, Fe, Mg, Sr, V, a rare earth element, or a combination thereof; D is O, F, S, P, or a combination thereof; G is Al, Cr, Mn, Fe, Mg, La, Ce, Sr, V, or a combination thereof; and L^{sup.1} is Mn, Al, or a combination thereof.

[0069] A positive electrode for a lithium secondary battery may include a current collector and a positive electrode active material layer formed on the current collector. The positive electrode active material layer may include a positive electrode active material and may further include a binder and/or a conductive material.

[0070] The content of the positive electrode active material is in a range of about 90 wt % to about 99.5 wt % on the basis of 100 wt % of the positive electrode active material layer, and the content of the binder and the conductive material is in a range of about 0.5 wt % to about 5 wt %, respectively, on the basis of 100 wt % of the positive electrode active material layer.

[0071] The current collector may be aluminum (Al) but is not limited thereto.

[0072] The negative electrode active material may include a material capable of reversibly intercalating/deintercalating lithium ions, lithium metal, an alloy of lithium metal, a material capable of being doped and undoped with lithium, or a transition metal oxide.

[0073] The material capable of reversibly intercalating/deintercalating lithium ions may be a carbon-based negative electrode active material, which may include, for example, crystalline carbon, amorphous carbon, or a combination thereof. Examples of the crystalline carbon may include graphite, such as natural graphite or artificial graphite, and examples of the amorphous carbon may include soft carbon, hard carbon, a pitch carbide, a meso-phase pitch carbide, sintered coke, and the like.

[0074] A Si-based negative electrode active material or a Sn-based negative electrode active material may be used as the material capable of being doped and undoped with lithium. The Si-based negative electrode active material may be silicon, a silicon-carbon composite, SiO_x ($0 < x < 2$), a Si-based alloy, or a combination thereof.

[0075] The silicon-carbon composite may be a composite of silicon and amorphous carbon. According to one embodiment, the silicon-carbon composite may be in the form of a silicon particle and amorphous carbon coated on the surface of the silicon particle.

[0076] The silicon-carbon composite may further include crystalline carbon. For example, the silicon-carbon composite may include a core including crystalline carbon and silicon particle and an amorphous carbon coating layer on the surface of the core.

[0077] A negative electrode for a lithium secondary battery may include a current collector and a negative electrode active material layer disposed on the current collector. The negative electrode active material layer may include a negative electrode active material and may further include a binder and/or a conductive material.

[0078] For example, the negative electrode active material layer may include about 90 wt % to about 99 wt % of a negative electrode active material, about 0.5 wt % to about 5 wt % of a binder, and about 0 wt % to about 5 wt % of a conductive material.

[0079] A non-aqueous binder, an aqueous binder, a dry binder, or a combination thereof may be used as the binder. When an aqueous binder is used as the negative electrode binder, a cellulose-based compound capable of imparting viscosity may be further included.

[0080] As the negative electrode current collector, one selected from copper foil, nickel foil, stainless steel foil, titanium foil, nickel foam, copper foam, conductive metal-coated polymer substrate, and combinations thereof may be used.

[0081] An electrolyte for a lithium secondary battery may include a non-aqueous organic solvent and a lithium salt.

[0082] The non-aqueous organic solvent acts as a medium through which ions involved in the electrochemical reaction of the battery can move.

[0083] The non-aqueous organic solvent may be a carbonate-based, an ester-based, an ether-based, a ketone-based, an alcohol-based solvent, an aprotic solvent, and may be used alone or in combination of two or more.

[0084] In addition, when a carbonate-based solvent is used, a mixture of cyclic carbonate and chain carbonate may be used.

[0085] Depending on the type of lithium secondary battery, a separator may be present between the first electrode plate (e.g., the negative electrode) and the second electrode plate (e.g., the positive electrode). As the separator, polyethylene, polypropylene, polyvinylidene fluoride, or a multilayer film of two or more layers thereof may be used.

[0086] The separator may include a porous substrate and a coating layer including an organic material, an inorganic material, or a combination thereof on one or both surfaces of the porous substrate.

[0087] The organic material may include a polyvinylidene fluoride-based heavy antibody or a (meth) acrylic polymer.

[0088] The inorganic material may include inorganic particles selected from Al_2O_3 , SiO_2 , TiO_2 , SnO_2 , CeO_2 , MgO , NiO , CaO , GaO , ZnO , ZrO_2 , Y_2O_3 , SrTiO_3 , BaTiO_3 , $\text{Mg}(\text{OH})_2$, boehmite, and combinations thereof but is not limited thereto.

[0089] The organic material and the inorganic material may be mixed in one coating layer or may be in the form of a coating layer containing an organic material and a coating layer containing an inorganic material that are laminated on each other.

[0090] FIG. 2A is a schematic diagram showing an exemplary apparatus **200** for manufacturing the secondary battery **100** according to some embodiments of the present disclosure. As shown in FIG.

2A, the exemplary apparatus **200** for manufacturing the secondary battery **100** according to the present disclosure may include an unwinding unit **210**, a fold prevention portion forming unit **220**, a notching unit **230**, and a winding unit **240**. In one or more embodiments, dancer rollers **251** for controlling tension of the electrode plate **111** may be further provided between the unwinding unit **210** and the fold prevention portion forming unit **220**. In one or more embodiments, dancer rollers **252** for controlling tension of the electrode plate **111** may be further provided between the notching unit **230** and the winding unit **240**. In one or more embodiments, various auxiliary rollers may be further provided on the travel path of the electrode plate **111** in order to prevent meandering of the electrode plate **111** and to control tension of the electrode plate **111**. However, description thereof will be omitted so as not to obscure the gist of the disclosure.

[0091] The unwinding unit **210** may unwind the electrode plate **111** wound on a first roller **211** at a predetermined speed. To this end, the first roller **211** may be coupled to an electric motor to rotate the first roller **211** at a predetermined speed.

[0092] In one or more embodiments, as described above, the electrode plate **111** may include a substrate portion **1111**, an active material coated portion **1112** in which an active material is coated on the substrate portion **1111**, and an uncoated portion **1115**, as shown in FIG. 3A, in which an active material is not coated on the substrate portion **1111**.

[0093] The fold prevention portion forming unit **220** may form the fold prevention portion **1114** on the uncoated portion **1115** of the electrode plate **111** unwound from the unwinding unit **210**. In one or more embodiments, the fold prevention portion forming unit **220** may include a driving roller **221** and a driven roller **222**. In one or more embodiments, the driving roller **221** may be rotated in a predetermined direction by an electric motor, and the driven roller **222** may be rotated in a predetermined direction by contact with the electrode plate **111** that travels while in contact with the driving roller **221**. In one or more embodiments, the driving roller **221** and/or the driven roller **222** may have an uneven pattern, an engraved pattern, and/or an embossed pattern in order to form and/or pattern the fold prevention portion **1114** on the uncoated portion **1115** of the electrode plate **111**.

[0094] The notching unit **230** may notch the uncoated portion **1115** of the electrode plate **111** received from the fold prevention portion forming unit **220** to form the substrate tab (or uncoated portion tab) **1113**. In one or more embodiments, the notching unit **230** may include a laser tool, a blade tool, and/or a punching tool.

[0095] The winding unit **240** may wind the electrode plate **111** received from the notching unit **230** on a second roller **241**. In one or more embodiments, the second roller **241** may be coupled to an electric motor to rotate at a predetermined speed to wind the electrode plate **111** including the substrate tab **1113** on which the fold prevention portion **1114** is formed.

[0096] According to some of embodiments the present disclosure, because the fold prevention portion **1114** is formed on the uncoated portion **1115** of the electrode plate **111** by the fold prevention portion forming unit **220**, it may be possible to prevent wrinkles or folding of the substrate tab **1113** and rupture of the electrode plate **111** during subsequent travel of the electrode plate **111**, notching of the electrode plate **111**, and/or winding of the electrode plate **111**.

[0097] FIG. 2B is a flowchart showing an exemplary method of manufacturing the secondary battery **100** according to some embodiments of the present disclosure. As shown in FIG. 2B, the exemplary method of manufacturing the secondary battery **100**, in some embodiments, may include a step of unwinding an electrode plate (S1), a step of forming a fold prevention portion (S2), a step of notching a substrate tab (S3), and a step of winding the electrode plate (S4). In one or more embodiments, the step of unwinding the electrode plate (S1) may be performed by the unwinding unit **210**, the step of forming the fold prevention portion (S2) may be performed by the fold prevention portion forming unit **220**, the step of notching the substrate tab (S3) may be performed by the notching unit **230**, and the step of winding the electrode plate (S4) may be performed by the winding unit **240**. Hereinafter, the above-described apparatus and method for manufacturing the

secondary battery will be described in more detail.

[0098] FIG. 3A is a perspective view showing the exemplary fold prevention portion forming unit **220** of the exemplary apparatus **200** for manufacturing the secondary battery **100** according to some embodiments of the present disclosure. FIG. 3B is a front view showing the driving roller **221** of the exemplary fold prevention portion forming unit **220** according to some embodiments of the present disclosure. FIG. 3C illustrates a side view and a partial enlarged view showing the driving roller **221** of the exemplary fold prevention portion forming unit **220** according to some embodiments of the present disclosure.

[0099] As shown in FIGS. 3A to 3C, the fold prevention portion forming unit **220** may include the driving roller **221** and the driven roller **222**. In one or more embodiments, the driving roller **221** may include a circular area **2211** and an uneven area **2212**, which are disposed on the same axis while being spaced apart from each other. In one or more embodiments, the driving roller **221** may include or be referred to as a polygonal roller. The driven roller **222** may rotate in close proximity to the uneven area **2212** of the driving roller **221**. In one or more embodiments, the driven roller **222** may include or be referred to as a urethane roller or a nip roller.

[0100] In one or more embodiments, the electrode plate **111** may travel between the driving roller **221** and the driven roller **222**. For example, the electrode plate **111** may travel between the driving roller **221** and the driven roller **222** in a state of being interposed therebetween (e.g., in contact therewith). In one or more embodiments, the active material coated portion **1112** of the electrode plate **111** may travel while in contact with the circular area **2211** of the driving roller **221**. In one or more embodiments, the uncoated portion **1115** of the electrode plate **111** may travel while in contact with the uneven area **2212** of the driving roller **221** and the driven roller **222**, such that the fold prevention portion **1114** may be imprinted on the uncoated portion **1115**. In one or more embodiments, the width of the active material coated portion **1112** may be similar or identical to the width of the circular area **2211** of the driving roller **221**. In one or more embodiments, the width of the uncoated portion **1115** may be similar or identical to the width of the uneven area **2212** of the driving roller **221** and/or the width of the driven roller **222**.

[0101] In one or more embodiments, the elastic modulus of the uneven area **2212** may be greater than that of the driven roller **222**. For example, the uneven area **2212** may be more difficult to deform than the driven roller **222**. The driven roller **222** may be easier to deform than the uneven area **2212**. In one or more embodiments, the hardness (e.g., Shore hardness) of the driven roller **222** may be about 30 A to about 50 A. The hardness of the uneven area **2212** may be greater than that of the driven roller **222**. In one or more embodiments, the uncoated portion **1115** may include a first surface and a second surface opposite the first surface. The uneven area **2212** may contact the first surface of the uncoated portion **1115**, and the driven roller **222** may contact the second surface of the uncoated portion **1115**. In one or more embodiments, the rotational axis of the uneven area **2212** and the rotational axis of the driven roller **222** may overlap each other in a vertical direction. The uneven area **2212** may press the surface of the driven roller **222** to a predetermined depth while deforming the uncoated portion **1115** so that the fold prevention portion **1114** is formed on the uncoated portion **1115**. In one or more embodiments, the surface of the driven roller **222** may be deformed when contacted/pressed by the uneven area **2212**, and may be restored to the original shape thereof when released from the uneven area **2212**. Thus, in some embodiments, the substrate portion **1111**, traveling in contact with the uneven area **2212** of the driving roller **221** and the driven roller **222**, may comprise the fold prevention portion **1114** having a predetermined shape without rupture of the electrode plate **111**. The shape of the uneven area **2212** may be transferred to the uncoated portion **1115** as is, thereby forming the fold prevention portion **1114**.

[0102] In one or more embodiments, the uneven area **2212** of the driving roller **221** may have a depth of about 0.1 mm to about 2 mm. Thus, in some embodiments, the fold prevention portion **1114** may also have a depth of about 0.1 mm to about 2 mm. In one or more embodiments, the uneven area **2212** of the driving roller **221** may have a pitch of about 1 mm to about 4 mm. Thus, in

some embodiments, the fold prevention portion **1114** may also have a pitch of about 1 mm to about 4 mm.

[0103] FIG. 4 is a schematic diagram showing the operation of the exemplary apparatus **200** for manufacturing the secondary battery **100** according to some embodiments of the present disclosure. As shown in FIG. 4, the fold prevention portion **1114** may be formed on the uncoated portion **1115** of the electrode plate **111** by the fold prevention portion forming unit **220**, and then the substrate tab **1113** may be formed on the uncoated portion **1115** by the notching unit **230**. In one or more embodiments, a part of the uncoated portion **1115** may be removed by the notching unit **230** and/or through a notching process, thereby forming the substrate tab **1113** that may be connected to a lead tab.

[0104] FIGS. 5A to 5D are schematic diagrams showing various forms of the fold prevention portion **1114** of the electrode plate **111** and the driving roller **221** of the exemplary apparatus **200** for manufacturing the secondary battery **100** according to the present disclosure.

[0105] As shown in FIG. 5A, a fold prevention portion **1114a** may be formed on the uncoated portion **1115** of the electrode plate **111** by the fold prevention portion forming unit **220**. The fold prevention portion **1114a** may be formed in a pattern including a plurality of straight lines extending in a direction substantially perpendicular to the longitudinal direction of the uncoated portion **1115**. In some embodiments, the uneven area **2212a** of the driving roller **221** of the fold prevention portion forming unit **220** may include a plurality of uneven portions extending in a direction substantially parallel to the rotational axis of the driving roller **221**.

[0106] As shown in FIG. 5B, a fold prevention portion **1114b** may be formed on the uncoated portion **1115** of the electrode plate **111** by the fold prevention portion forming unit **220**. The fold prevention portion **1114b** may be formed in a pattern including a plurality of straight lines extending in a direction substantially parallel to the longitudinal direction of the uncoated portion **1115**. In some embodiments, the uneven area **2212b** of the driving roller **221** of the fold prevention portion forming unit **220** may include a plurality of uneven portions extending in a direction substantially perpendicular to the rotational axis of the driving roller **221**.

[0107] As shown in FIG. 5C, a fold prevention portion **1114c** may be formed on the uncoated portion **1115** of the electrode plate **111** by the fold prevention portion forming unit **220**. The fold prevention portion **1114c** may be formed in a pattern including a plurality of oblique lines extending at a predetermined angle with respect to the longitudinal direction of the uncoated portion **1115**. In some embodiments, the uneven area **2212c** of the driving roller **221** of the fold prevention portion forming unit **220** may include a plurality of uneven portions extending at a predetermined angle with respect to the rotational axis of the driving roller **221**.

[0108] As shown in FIG. 5D, a fold prevention portion **1114d** may be formed on the uncoated portion **1115** of the electrode plate **111** by the fold prevention portion forming unit **220**. The fold prevention portion **1114d** may be formed in a pattern including a plurality of straight lines extending in a direction substantially perpendicular to the longitudinal direction of the uncoated portion **1115** and at least one straight line extending in a direction substantially parallel to the longitudinal direction of the uncoated portion **1115**. In some embodiments, the uneven area **2212d** of the driving roller **221** of the fold prevention portion forming unit **220** may include a plurality of uneven portions extending in a direction substantially parallel to the rotational axis of the driving roller **221** and at least one uneven portion extending in a direction substantially perpendicular to the rotational axis of the driving roller **221**.

[0109] FIGS. 6A and 6B show a battery pack **300** according to one or more embodiments of the present disclosure. The battery pack **300** may include a plurality of battery modules **200** and a housing **310** for accommodating the plurality of battery modules **200**. For example, the housing **310** may include first and second housings **311** and **312** coupled in opposite directions through the plurality of battery modules **200**. The plurality of battery modules **200** may be electrically connected to each other by using a bus bar **251**, and the plurality of battery modules **200** may be

electrically connected to each other in a series/parallel or series-parallel mixed method, thereby obtaining desired (e.g., required) electrical output. In the drawing, for convenience of illustration, parts such as bus bars, cooling units, and external terminals for electrical connection of battery cells are omitted. In one or more embodiments, battery pack **300** may be mounted in a vehicle. The vehicle may be, for example, an electric vehicle, a hybrid vehicle, or a plug-in hybrid vehicle. A vehicle may include a four-wheeled vehicle or a two-wheeled vehicle.

[0110] FIGS. 7A and 7B show vehicle body parts **400** and vehicle **500** according to one or more embodiments of the present disclosure including the battery pack **300** shown in FIGS. 6A and 6B.

[0111] In FIG. 7A, a battery pack **300** may include a battery pack cover **311**, which is a part of a vehicle underbody **410** and may correspond to the first housing, and a pack frame **312**, which is disposed under the vehicle underbody **410** and may correspond to the second housing. The battery pack cover **311** and the pack frame **312** may be integrally formed with a vehicle floor **420**. The vehicle underbody **410** separates the inside and outside of a vehicle, and the pack frame **312** may be disposed outside the vehicle.

[0112] In FIG. 7B, a vehicle **500** may be formed by combining additional parts, such as a hood **510** in front of the vehicle **500** and fenders **520** respectively located in the front and rear of the vehicle **500** to a vehicle body part **400**. The vehicle **500** may include the battery pack **300** including the battery pack cover **311** and the pack frame **312**, and the battery pack **300** may be coupled to the vehicle body part **400**.

[0113] As is apparent from the above description, the present disclosure may provide a secondary battery manufacturing apparatus and method capable of preventing wrinkles and folding of a substrate tab during a process of manufacturing an electrode plate of a secondary battery and preventing rupture of the electrode plate during travel or winding of the electrode plate and a secondary battery manufactured thereby.

[0114] Although the present disclosure has been described with reference to embodiments and drawings illustrating aspects thereof, the present disclosure is not limited thereto. Various modifications and variations can be made by a person skilled in the art to which the present disclosure belongs within the scope of the technical spirit of the present disclosure and the claims and their equivalents, below.

Claims

1. An apparatus for manufacturing a secondary battery, the apparatus comprising: an unwinding unit configured to unwind an electrode plate wound on a first roller, the electrode plate comprising an active material coated portion in which an active material is coated on a substrate portion and an uncoated portion in which the active material is not coated on the substrate portion; a fold prevention portion forming unit configured to form a fold prevention portion on the uncoated portion of the electrode plate; a notching unit configured to notch the uncoated portion comprising the fold prevention portion formed thereon to form a substrate tab; and a winding unit configured to wind the electrode plate comprising the substrate tab formed thereon on a second roller.
2. The apparatus as claimed in claim 1, wherein the fold prevention portion forming unit comprises: a driving roller comprising a circular area and an uneven area, the circular area and the uneven area disposed on a same axis while being spaced apart from each other; and a driven roller configured to rotate in close proximity to the uneven area of the driving roller, wherein the electrode plate travels between the driving roller and the driven roller such that the active material coated portion of the electrode plate travels in contact with the circular area of the driving roller and the uncoated portion of the electrode plate travels in contact with the uneven area of the driving roller and the driven roller, such that the fold prevention portion is imprinted on the uncoated portion.
3. The apparatus as claimed in claim 2, wherein an elastic modulus of the uneven area is greater

than an elastic modulus of the driven roller.

4. The apparatus as claimed in claim 2, wherein the uncoated portion comprises a first surface and a second surface opposite the first surface, wherein the uneven area contacts the first surface of the uncoated portion, and the driven roller contacts the second surface of the uncoated portion, and wherein a rotational axis of the uneven area and a rotational axis of the driven roller overlap each other in a vertical direction.

5. The apparatus as claimed in claim 1, wherein the fold prevention portion comprises a pattern comprising a plurality of straight lines extending in a direction perpendicular to a longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

6. The apparatus as claimed in claim 1, wherein the fold prevention portion comprises a pattern comprising a plurality of straight lines extending in a direction parallel to a longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

7. The apparatus as claimed in claim 1, wherein the fold prevention portion comprises a pattern comprising a plurality of oblique lines extending at an angle with respect to a longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

8. The apparatus as claimed in claim 1, wherein the fold prevention portion comprises a pattern comprising a plurality of straight lines extending in a direction perpendicular to a longitudinal direction of the uncoated portion and at least one straight line extending in a direction parallel to the longitudinal direction of the uncoated portion by the fold prevention portion forming unit.

9. The apparatus as claimed in claim 2, wherein the uneven area of the driving roller has a depth of 0.1 mm to 2 mm.

10. A method of manufacturing a secondary battery, the method comprising: unwinding an electrode plate wound on a first roller, the electrode plate comprising an active material coated portion in which an active material is coated on a substrate portion and an uncoated portion in which an active material is not coated on the substrate portion; forming a fold prevention portion on the uncoated portion of the electrode plate; notching the uncoated portion comprising the fold prevention portion formed thereon to form a substrate tab; and winding the electrode plate comprising the substrate tab formed thereon on a second roller.

11. The method as claimed in claim 10, wherein the forming a fold prevention portion comprises providing a driving roller comprising a circular area and an uneven area, the circular area and the uneven area disposed on a same axis while being spaced apart from each other, and a driven roller configured to rotate in close proximity to the uneven area of the driving roller, and wherein the electrode plate travels between the driving roller and the driven roller such that the active material coated portion of the electrode plate travels in contact with the circular area of the driving roller and the uncoated portion of the electrode plate travels in contact with the uneven area of the driving roller and the driven roller, such that the fold prevention portion is imprinted on the uncoated portion.

12. The method as claimed in claim 11, wherein an elastic modulus of the uneven area is greater than an elastic modulus of the driven roller.

13. The method as claimed in claim 11, wherein the uncoated portion comprises a first surface and a second surface opposite the first surface, wherein the uneven area contacts the first surface of the uncoated portion, and the driven roller contacts the second surface of the uncoated portion, and wherein a rotational axis of the uneven area and a rotational axis of the driven roller overlap each other in a vertical direction.

14. The method as claimed in claim 10, wherein, in the forming a fold prevention portion, the fold prevention portion is formed in a pattern comprising a plurality of straight lines extending in a direction perpendicular to a longitudinal direction of the uncoated portion.

15. The method as claimed in claim 10, wherein, in the forming a fold prevention portion, the fold prevention portion is formed in a pattern comprising a plurality of straight lines extending in a direction parallel to a longitudinal direction of the uncoated portion.

- 16.** The method as claimed in claim 10, wherein, in the forming a fold prevention portion, the fold prevention portion is formed in a pattern comprising a plurality of oblique lines extending at an angle with respect to a longitudinal direction of the uncoated portion.
- 17.** The method as claimed in claim 10, wherein, in the forming a fold prevention portion, the fold prevention portion is formed in a pattern comprising a plurality of straight lines extending in a direction perpendicular to a longitudinal direction of the uncoated portion and at least one straight line extending in a direction parallel to the longitudinal direction of the uncoated portion.
- 18.** The method as claimed in claim 11, wherein the uneven area of the driving roller is formed to have a depth of 0.1 mm to 2 mm.
- 19.** A secondary battery manufactured by the method as claimed in claim 10.
- 20.** An apparatus for manufacturing a secondary battery, the apparatus comprising: a fold prevention portion forming unit configured to form a fold prevention portion on an uncoated portion of an electrode plate; and a notching unit configured to notch the uncoated portion comprising the fold prevention portion formed thereon to form a substrate tab.
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